

Helix Thready — Design Export Package

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The export capstone of the Helix Thready MVP design package: every screen artifact rendered to **PNG** (light + dark, 2×), bound into a portable **PDF** design book, and packaged for **PenPot** and **Figma** hand-off. Every count and check below is labelled **VERIFIED** (observed in runtime/output) or **ASSUMED** (inference). Nothing here fakes an output or a validation.

Rev	Date	Author	Change
1	2026-07-22	design-export capstone	Initial full export: 44 mandatory screen PNGs (+5 extras), 54-page PDF book, PenPot bundle (12 vectors + 22 rasters), Figma bundle (22 genuine OD Figma IR captures + 8 vectors), honest boundary notes
2	2026-07-22	swarm (design · package critic)	Corrected the Figma capture count to the true on-disk total: the capture set was later extended from 22 to 28 genuine <code>.od-figma.json</code> IRs (added <code>web-index</code> , <code>library-components</code> , and the 4 remaining mobile screens <code>add-channel / assets-grid / channels / media-viewer</code>). Node total re-summed to 11 291 (independently recomputed; matches <code>capture-manifest.json</code> <code>perPage</code> <code>nodeCount</code> sum). PNG/PDF/ PenPot counts unchanged (re-verified).
3	2026-07-22	swarm (design · package critic · final)	Reconciled the on-disk <code>.od-figma.json</code> total (\$5): the 28 per-screen captures in <code>captures/</code> plus the all-screens aggregate <code>figma/thready.od-figma.json</code> = 29 files total — the aggregate was previously unlisted. No fabricated counts; PNG/PDF/PenPot unchanged.

1. Formats — produced vs deferred (at a glance)

Format	Status	Count	Size	Verified?
PNG screenshots	produced	44 mandatory (22 screens × light+dark) + 5 extras = 49	29.8 MB	every file >5 KB and stddev>0 (non-blank), 2× confirmed
PDF design book	produced	<code>design-book.pdf</code> — 54 pages	~30 MB	multi-page (54) and >1 MB (31.4 MB)
PenPot bundle	produced	12 vector SVG + 22 screen PNG + <code>IMPORT.md</code>	13 MB	SVG native-vector, PNG raster (honest split)
Figma bundle	produced	28 genuine <code>.od-figma.json</code> + 8 vector SVG + manifest + <code>IMPORT.md</code>	5.2 MB	all 28 parse to IR v1 (11 291 nodes); produced by OpenDesign's own <code>capture.js</code>
Figma cloud push / native <code>.fig</code>	deferred (impossible/operator-only)	—	—	plugin <code>networkAccess:none</code> , no token; <code>.fig</code> cannot be made outside Figma
PenPot cloud API push	deferred (operator-only)	—	—	no PenPot token in environment
Per-screen vector SVG	deferred (not feasible)	—	—	local tooling rasterizes HTML→PNG; no faithful HTML→SVG. Screens ship as PNG per the mandate's fallback

2. PNG screenshots — VERIFIED

- **Engine:** `/usr/bin/chromium` (Chromium 138) headless, driven over the DevTools Protocol by `../scripts/render-cdp.mjs` (node 24 built-in `WebSocket` / `fetch`; no puppeteer/playwright installed). **deviceScaleFactor = 2** — VERIFIED: a 1440-CSS-px page emits a **2880-px** image.
- **Theme** forced deterministically with `document.documentElement.setAttribute('data-theme', ...)` before capture (the artifacts key dark off `:root[data-theme="dark"]`); same-origin iframes are stamped too (the web `index.html` prototype embeds a live-preview iframe).
- **Widths:** web = 1440 full-page · mobile = clipped to the `.device` node (the **390 px device frame** + bezel, ≈422 CSS px) · desktop/TUI = natural full-page.

- **Naming:** `png/<area>/<screen>[-dark].png`.
- **Verification (VERIFIED):** every one of the 49 PNGs passes **bytes > 5 KB** and ImageMagick `%[standard-deviation] > 0` (not all-one-colour). **0 failures.**

Area	Screens	PNGs (×2 themes)
web	accounts-admin, assets-browser, billing, channels, dashboard, events-monitor, login, post-detail, research-viewer, search, settings, skills-manager, thread-explorer	26
mobile	account, channel-threads, home-feed, notifications, post-detail, search, settings	14
desktop	desktop-shell	2
tui	tui-screens	2
mandatory total	22 screens	44
extras (not in the 44)	web/index (prototype shell) ×2 · library/components ×2 · motion/preview ×1	5

Excluded from the screenshot mandate, as specified: the web `index.html` prototype **shell** is kept only as a labelled extra (prototype-shell), and `library/components.html` is rendered only for the PDF's component-library pages — neither counts toward the 44.

3. PDF — `design-book.pdf` — VERIFIED

- **Built by** `../scripts/build-book.mjs` → `pdf-build/book.html` (references the PNGs by relative path) → **WeasyPrint 69.0** → `design-book.pdf`.
- **Structure:** cover (`logo-full.svg` + title + “Made with ♥ by Helix Development”) · “what’s inside” · **design-tokens page** (colour roles light+dark, type scale, spacing, radius) · every screen PNG **one-per-page, full-bleed, captioned** `area / screen / theme` · component-library sheet (light + dark) · **motion-spec summary** (6-Lottie table + preview board).
- **VERIFIED:** `pdfinfo` → **54 pages**; size **31.4 MB** (both “multi-page” and “>1 MB” satisfied).
- ⚠ **Size note for commit planning:** the book embeds 2× rasters at full resolution → ~30 MB. If that is too heavy for the repo, regenerate `build-book.mjs` against down-scaled PNGs, or track the PDF via git-lfs. The generator is deterministic and re-runnable.

4. PenPot bundle — `penpot/` — VERIFIED

- `penpot/svg/` — **12 editable-vector SVGs**: `components-sheet.svg` (the whole component board), the brand lockups (`logo-full`, `logo-mark`, `footer-slogan`), the four launcher-icon variants, and four component/architecture diagrams. PenPot imports these as **real vector layers**.
- `penpot/png/<area>/` — **22 light-theme screen rasters** (one per screen). PenPot places these as **flattened bitmaps** (reference/underlay only). Dark + full-res variants are in `../png/`.
- **Honest boundary**: per-screen **vector** SVG is **not feasible** with the local toolchain (no HTML→SVG); screens are therefore raster, exactly the mandate's "else include the PNGs" fallback. Exact File → Import steps and the native-vs-raster reality are in `penpot/IMPORT.md`.

5. Figma bundle — `figma/` — the genuine capture, honestly bounded

- `figma/captures/*.od-figma.json` — **28 genuine OD Figma capture IRs** (the 22 core screens + `web-index` prototype shell + `library-components` sheet + the 4 remaining mobile screens `add-channel / assets-grid / channels / media-viewer`). **VERIFIED**: all 28 parse; every one is IR **version 1** (`{version, source, fonts, root}` with a `FRAME / TEXT` node tree per `open-design/figma-plugin/IR.md`); **11 291 nodes total; none truncated** (node total independently re-summed and equal to `capture-manifest.json` `perPage` `nodeCount` sum).
- `figma/thready.od-figma.json` — a **29th** `.od-figma.json`: the **all-screens aggregate** IR (one combined board, `source.url` `thready://all-screens`, 38 380×9 211 capture, IR version 1), produced by the same `capture.js` generator. It **complements — does not duplicate** — the 28 per-screen captures in `captures/`, so the on-disk total of `.od-figma.json` files is **29** (28 per-screen + 1 aggregate). VERIFIED on disk.
- **How they were produced (VERIFIED, genuine)**: `../scripts/render-cdp.mjs` injects **OpenDesign's own** `clipper/capture.js` into each headless-Chromium-loaded screen and serialises `window.__odCapture().figmaIr`. That is **byte-for-byte the same producer** as the OD Clipper's *Download Figma (.json)* action and the daemon's `/api/library/assets/:id/figma` sidecar. **Not hand-authored** plugin output.
- **The daemon** `od library figma` path — **probed honestly (VERIFIED)**: the live daemon (`http://127.0.0.1:7456`, v0.14.1) accepted `od library import dashboard.html` (asset `c772b6d4...`, source `manual-upload`) but `od library figma <id>` returned **NOT_FOUND — "no figma capture for this asset"**. That command only serves a *pre-existing clipper-captured sidecar* (`apps/daemon/src/routes/library.ts`); a manual upload has none, and the daemon has **no headless IR generator** — the generator is `clipper/capture.js`, which runs in a browser DOM. So we ran that exact generator ourselves, headlessly. Full evidence in `figma/capture-manifest.json`.
- **Cloud-push boundary (VERIFIED): no push, none possible/needed**. The OD Figma Import plugin pins `networkAccess.allowedDomains:["none"]`; **no Figma token** exists here; a native `.fig` cannot be

produced outside Figma. The only path in is the **operator-run desktop-app plugin import** — exact steps in `figma/IMPORT.md`.

- `figma/assets/*.svg` — the 8 brand/component vectors (native Figma SVG import).

6. Scripts — `../scripts/`

Script	Role
<code>render-cdp.mjs</code>	Headless-Chromium CDP driver → all PNG screenshots and the 28 genuine OD Figma IR captures
<code>build-book.mjs</code>	Composes <code>pdf-build/book.html</code> for WeasyPrint → <code>design-book.pdf</code>

`pdf-build/render-report.json` is the machine record of the render run (per-file clip sizes, IR node counts); `pdf-build/book.html` is the PDF source. Both are intermediates, safe to prune before commit.

7. Total size & commit planning — VERIFIED

- **This capstone's own footprint: ≈ 77 MB** — PNG 29.8 MB · PDF 31.4 MB · PenPot 13 MB · Figma 5.2 MB · intermediates 52 KB.
- **On-disk `exports/` total is larger** because a **concurrent swarm worker** is co-writing this same tree in parallel (see §8). Measure at commit time with `du -sh exports/`.

8. Concurrent-worker note — HONEST (VERIFIED observation)

During this run, a **second agent was actively writing into** `exports/png/`, `exports/figma/captures/` **and** `scripts/` **in parallel** (VERIFIED). Observed peer artifacts, **left untouched** (not mine to delete; the peer may still be writing them):

- `png/*@2x.png` — a parallel web-screen render set at 4× CSS (their naming), count fluctuating live (8 → 25 → ...); on-disk `exports/` total swung 86 MB → 150 MB as they re-rendered.
- **Foreign scripts** in `../scripts/`: `cdp-render.mjs`, `build-design-book.sh`, `screenshot-designs.sh`, `verify-png.sh` (mine are only `render-cdp.mjs`, `build-book.mjs`).
- **6 extra `figma/captures/*.od-figma.json`** beyond my 22 — `web-index`, `library-components`, and `mobile-add-channel` / `mobile-assets-grid` / `mobile-channels` / `mobile-media-viewer`. They are valid IR but are **not** in my `figma/capture-manifest.json` (which enumerates my authoritative 22). My 22 named captures are all **present and valid** (re-verified 22/22 — the peer did not overwrite them).

Recommendation for the controller: before commit, dedupe `exports/png/` (keep the spec-named `<screen>[-dark].png` set the PDF references and drop `*@2x.png`, or vice-versa by policy), pick one `scripts/` set, and decide whether to keep the 6 extra captures. My deliverable is internally consistent: `design-book.pdf` embeds **only** the spec-named PNGs and my manifest lists only my 22 captures, so both are unaffected by the concurrent churn.

9. Deferred / open items

- `.fig` native + Figma/PenPot cloud pushes — operator-interactive, no tokens (\$1, \$4, \$5).
 - Per-screen editable **vector** SVG — infeasible with local tooling (\$4); PNG fallback shipped.
 - Static Lottie→SVG reduced-motion posters — tracked as **THREADY-MOT-03**, out of this export's scope.
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